

Simple Linear Regression

Statistical Methods Dossier with Subject-Specific Examples

1. Method Description

Simple linear regression models the linear relationship between one independent variable (X) and one dependent variable (Y).

Model: $Y = b_0 + b_1 \cdot X + e$, where b_0 is the intercept, b_1 is the slope, and e is the error term.

The goal is to find the best-fitting line that minimizes the sum of squared residuals (Ordinary Least Squares, OLS).

The coefficient b_1 indicates the expected change in Y when X increases by one unit.

R-squared (R^2) measures the proportion of variance in Y explained by X.

2. Subject-Specific Examples

Social Sciences: Does hours studied predict exam performance? IV: study hours, DV: exam score (0-100). Interpretation: Each additional study hour increases the expected exam score by 3.2 points ($p < .001$, $R^2 = .34$).

Economics: Does advertising expenditure predict sales? IV: ad spend (EUR), DV: monthly sales (EUR). Interpretation: Each additional EUR 1000 in ads increases sales by EUR 12,500.

Human Sciences: Does weekly therapy session count predict symptom reduction? IV: sessions per week, DV: BDI score reduction.

Humanities: Does text length (words) predict readability score? IV: word count, DV: Flesch-Reading-Ease index.

3. Important Concepts and Assumptions

Linearity: The relationship between X and Y must be linear (check scatterplot).

Independence of errors: Observations must be independent (Durbin-Watson test).

Homoscedasticity: Residual variance must be constant across X (Breusch-Pagan test, residuals-vs.-fitted plot).

Normality of residuals: Residuals should be approximately normally distributed (Q-Q plot, Shapiro-Wilk test).

No influential outliers: No single case should unduly influence the model (Cook's distance < 1).

4. Interpretation and practice

Report: 'Simple linear regression showed that

Effect size: R^2 of .01 (small), .09 (medium), .25 (large) per Cohen's benchmarks.

Standardized coefficient beta: expressed in SD units, comparable across predictors.

Prediction: Use the regression equation to predict Y for new values of X, with prediction intervals.

Software: R (lm), Python (statsmodels, sklearn.linear_model), SPSS (REGRESSION).

5. Code Examples

R: `fit <- lm(y ~ x, data = d); summary(fit); plot(fit)`

Python: `import statsmodels.api as sm; model = sm.OLS(y, sm.add_constant(X)).fit(); print(model.summary())`

SPSS: REGRESSION /DEPENDENT y /METHOD=ENTER x /SCATTERPLOT=(*ZRESID, *ZPRED).

Python (sklearn): `from sklearn.linear_model import LinearRegression; model = LinearRegression().fit(X, y)`